LAPORAN PRAKTIKUM ASD

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Sistem Informasi Bisnis / 1E

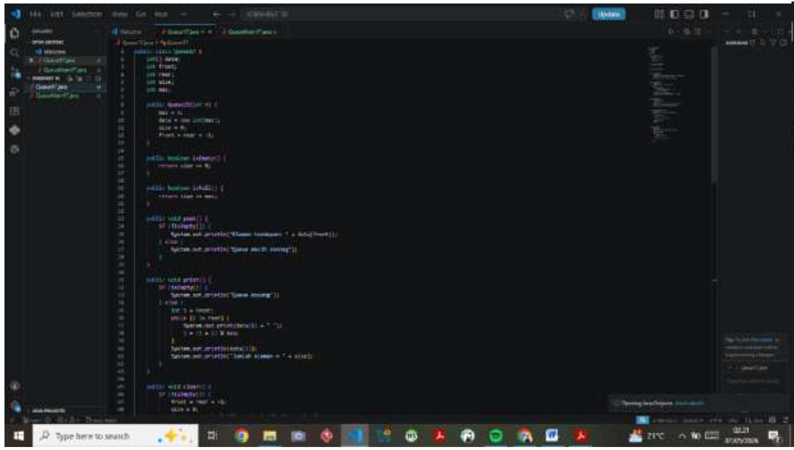
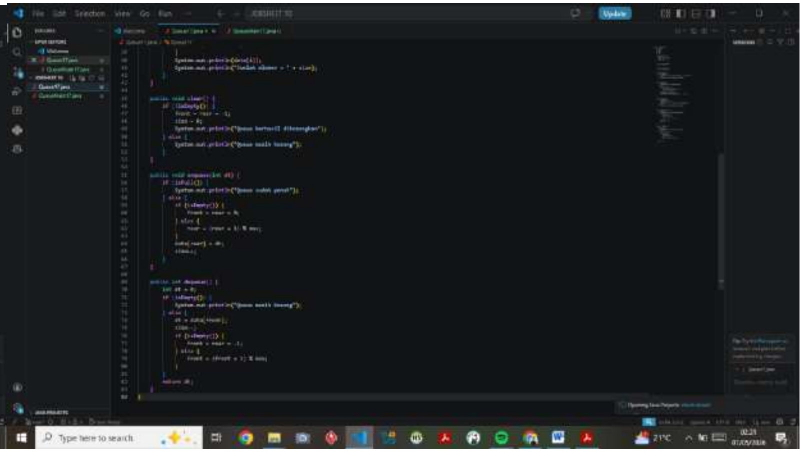
JOBSHEET 10 QUEUE

Tujuan Praktikum

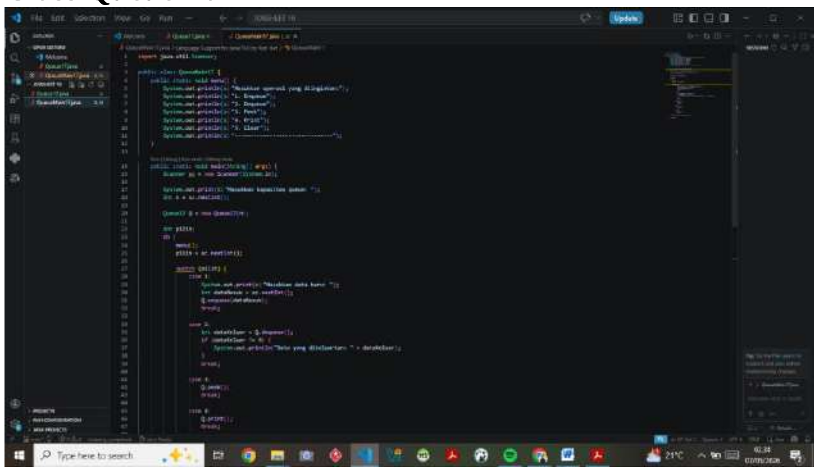
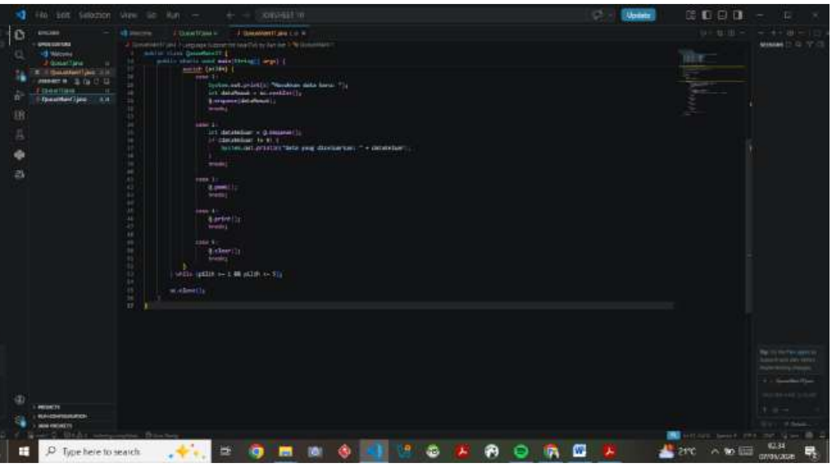
1. Mengenal struktur data Queue.
2. Membuat dan mendeklarasikan struktur data Queue.
3. Menerapkan algoritma Queue menggunakan array.

Praktikum 1 – Operasi Dasar Queue

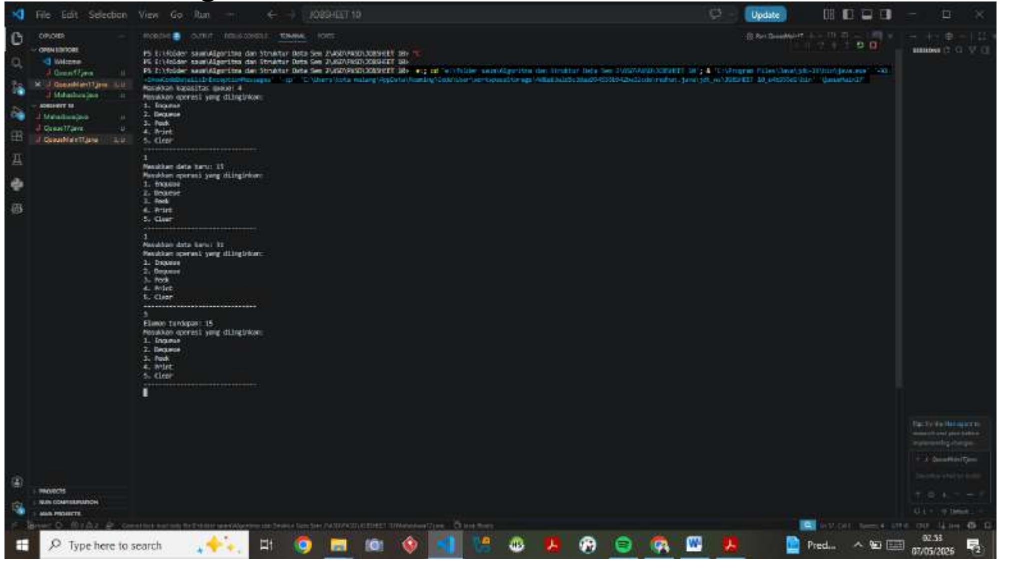
Class Queue

Class QueueMain

Hasil Run Program



Penjelasan Program

Program ini mengimplementasikan struktur data Queue (antrian) menggunakan array.

Method yang digunakan:

- Enqueue() → menambah data ke belakang queue.
- Dequeue() → menghapus data dari depan queue.
- peek() → melihat elemen terdepan.
- print() → menampilkan seluruh isi queue.
- clear() → menghapus semua isi queue.
- IsEmpty() → cek queue kosong.
- IsFull() → cek queue penuh.

Queue memakai konsep FIFO (First In First Out), artinya data yang masuk pertama akan keluar pertama.

Jawaban Pertanyaan Praktikum 1

1. Mengapa front dan rear = -1, size = 0?

Karena queue awalnya kosong.

- `front = -1` artinya belum ada elemen depan.
- `rear = -1` artinya belum ada elemen belakang.

- `size = 0` karena jumlah elemen masih nol.

---

2. Jelaskan potongan kode Enqueue

```
if (rear == max - 1) {
    rear = 0;
} else {
    rear++;
}
```

Kode ini digunakan untuk **circular queue**.
Jika rear sudah di indeks terakhir array, maka kembali ke indeks 0 agar array bisa dipakai ulang.

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3. Jelaskan potongan kode Dequeue

```
if (front == max - 1) {
    front = 0;
} else {
    front++;
}
```

Digunakan agar front berpindah maju setelah data dihapus.
Jika front sudah di indeks terakhir, maka kembali ke indeks 0.

---

4. Mengapa print mulai dari front?

Karena queue menampilkan data dari elemen pertama yang aktif.
Kalau mulai dari 0, bisa ikut menampilkan data lama yang sudah tidak termasuk queue.

---

5. Jelaskan kode berikut

```
i = (i + 1) % max;
```

Digunakan agar indeks bergerak circular.
Contoh:
max = 5
i = 4
Maka:
(4+1)%5 = 0
Jadi kembali ke awal array.

---

6. Potongan kode queue overflow

```
if (IsFull()) {
    System.out.println("Queue sudah penuh");
}
```

Overflow terjadi saat queue penuh tetapi masih ditambah data.

---

7. Modifikasi agar program berhenti saat overflow/underflow

Tambahkan:

```
System.exit(0);
```

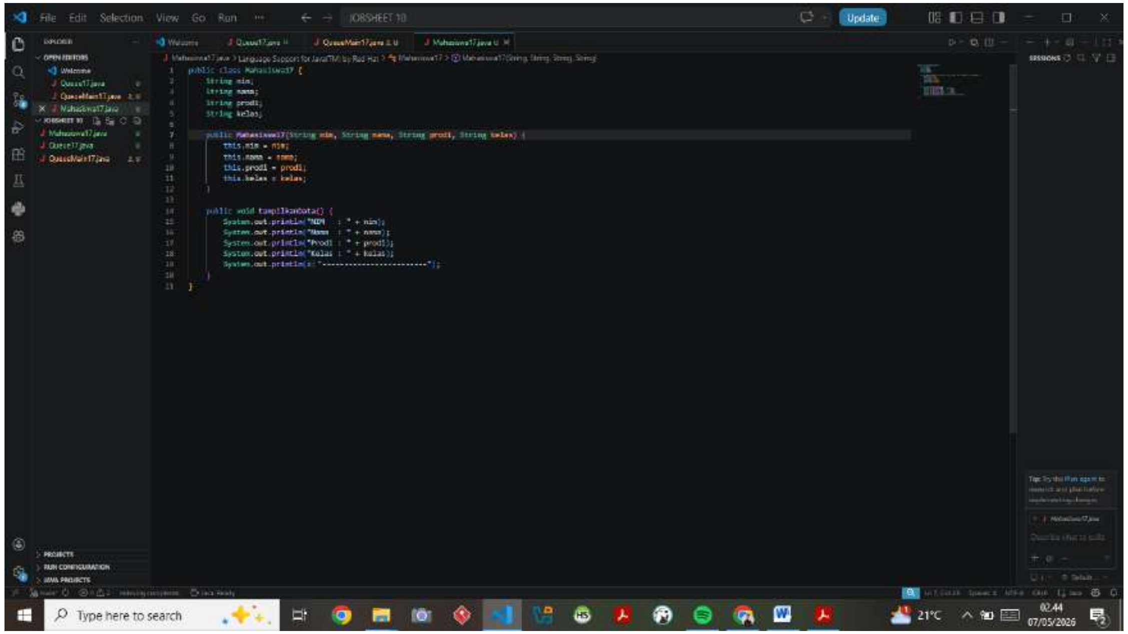
Pada kondisi:

queue penuh

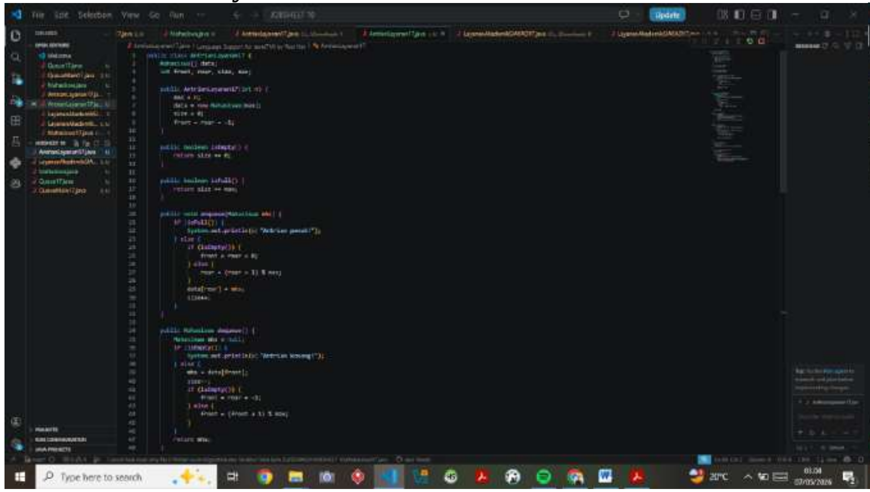
queue kosong saat dequeue

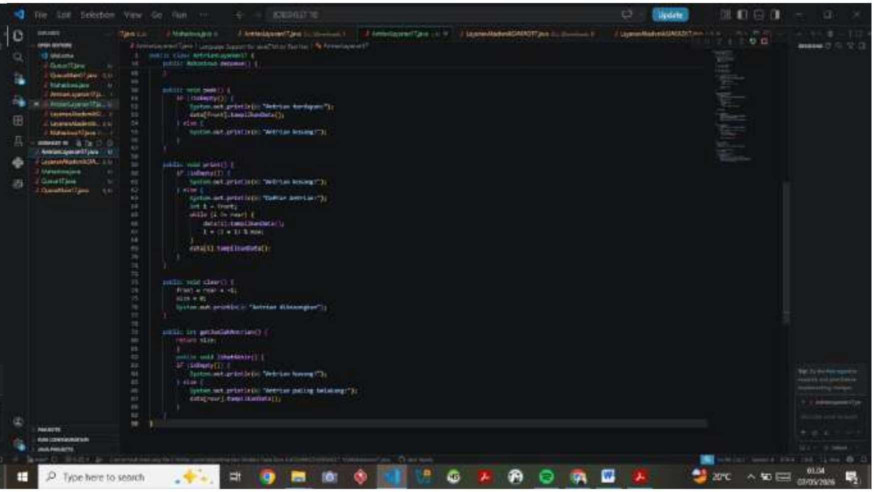
Praktikum 2 – Antrian Layanan Akademik

Class Mahasiswa

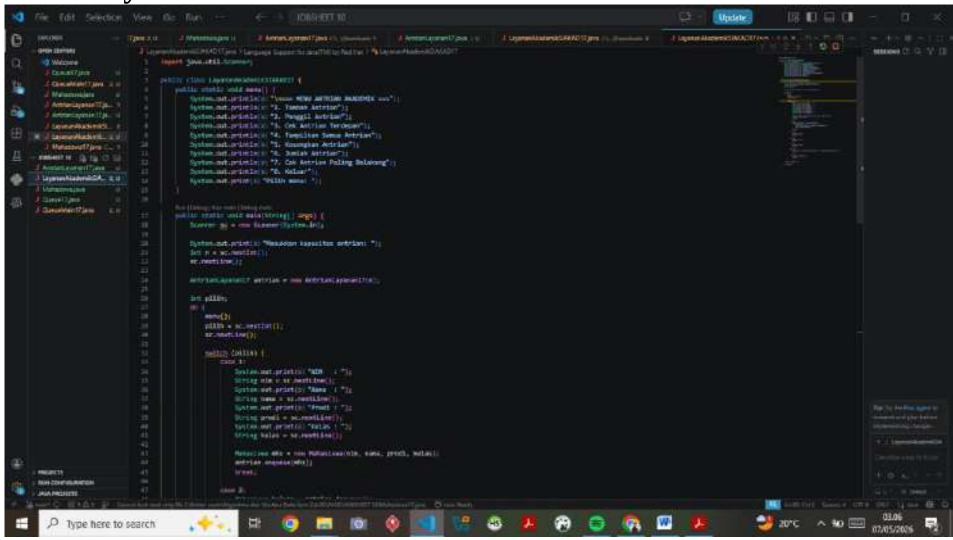


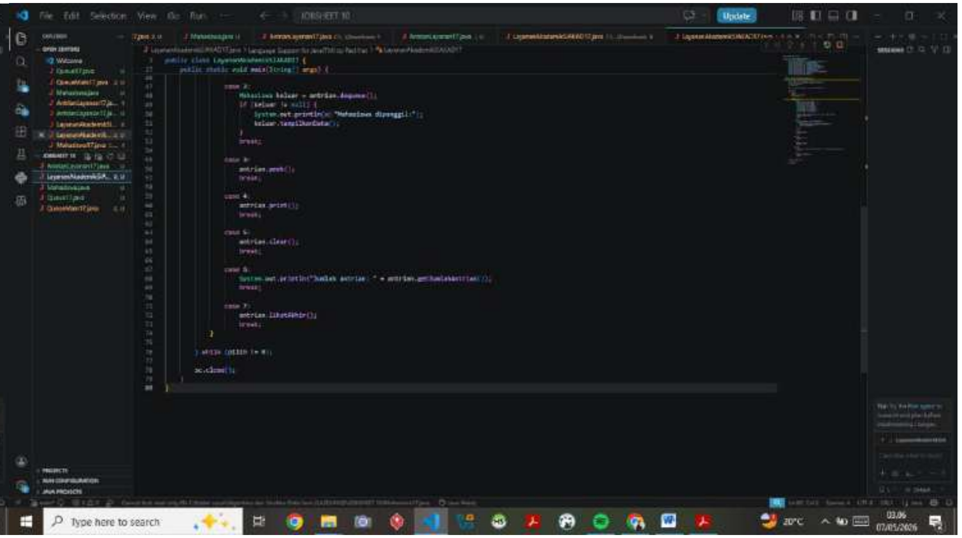
Class AntrianLayanan



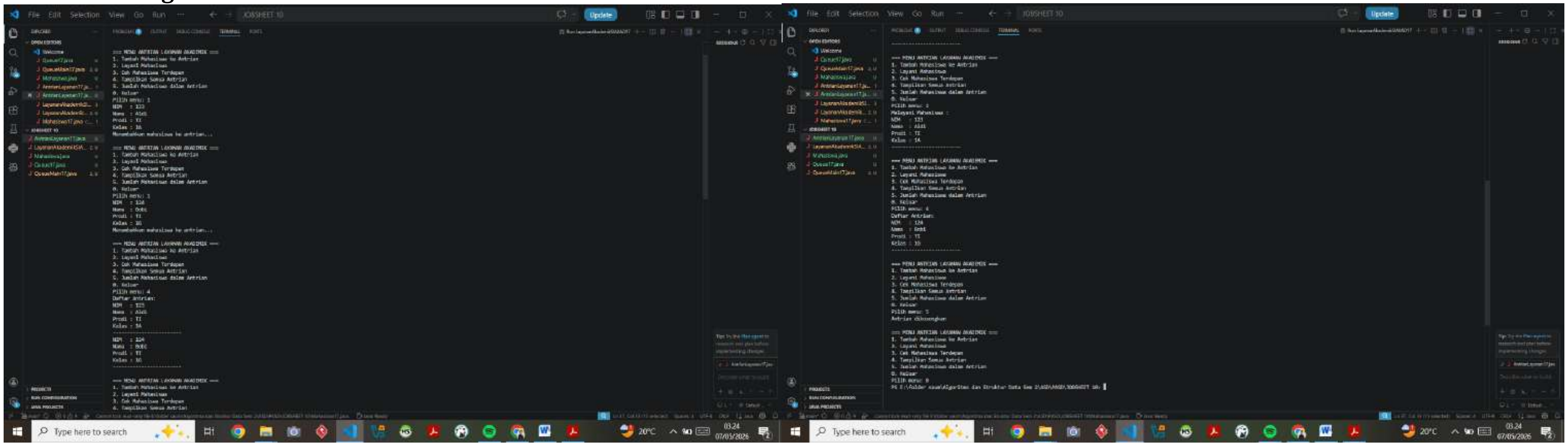


Class LayananAkademikSIAKAD





Hasil Run Program

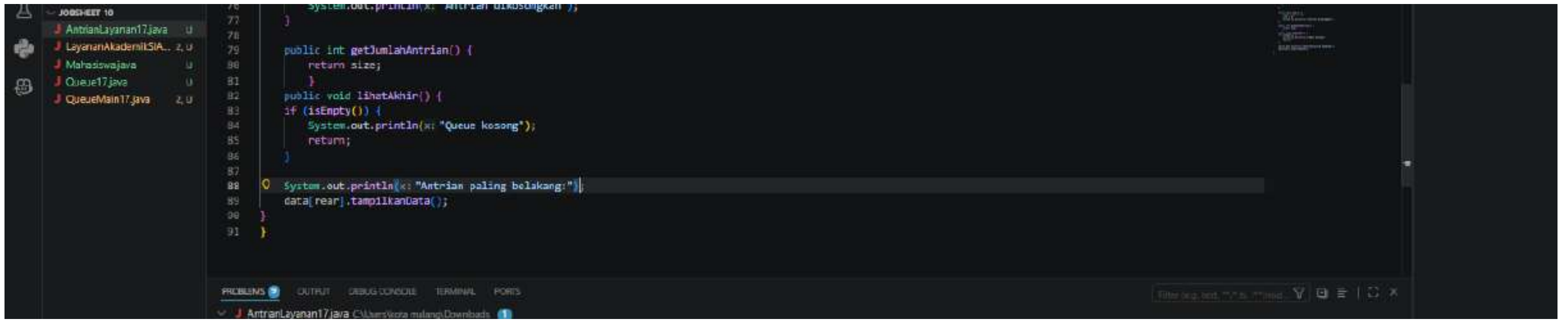


4.3 Jawaban Pertanyaan Praktikum 2

Tambahkan method LihatAkhir

Sudah ditambahkan:

lihatAkhir ()

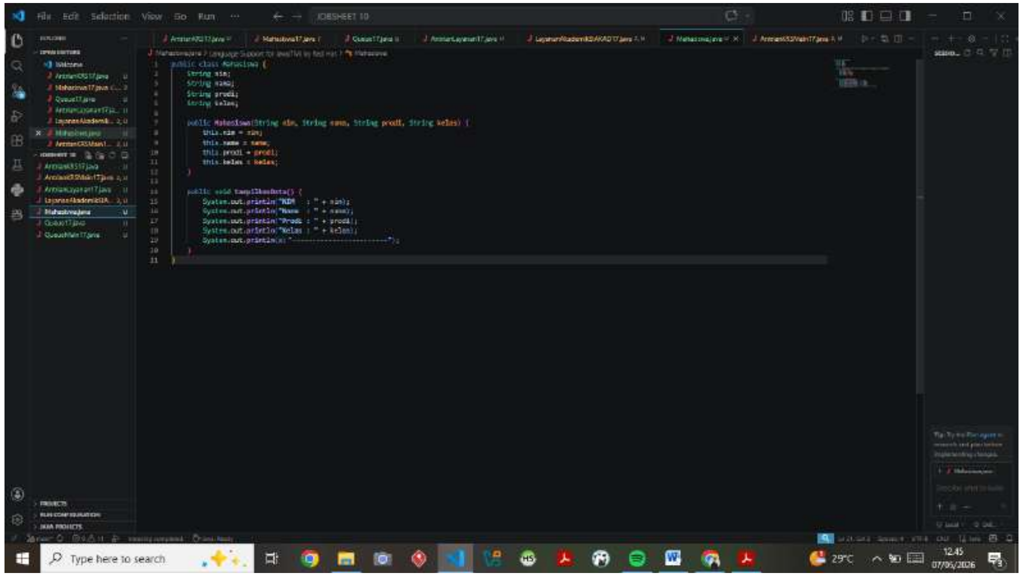


Menu:
6. Cek antrian belakang

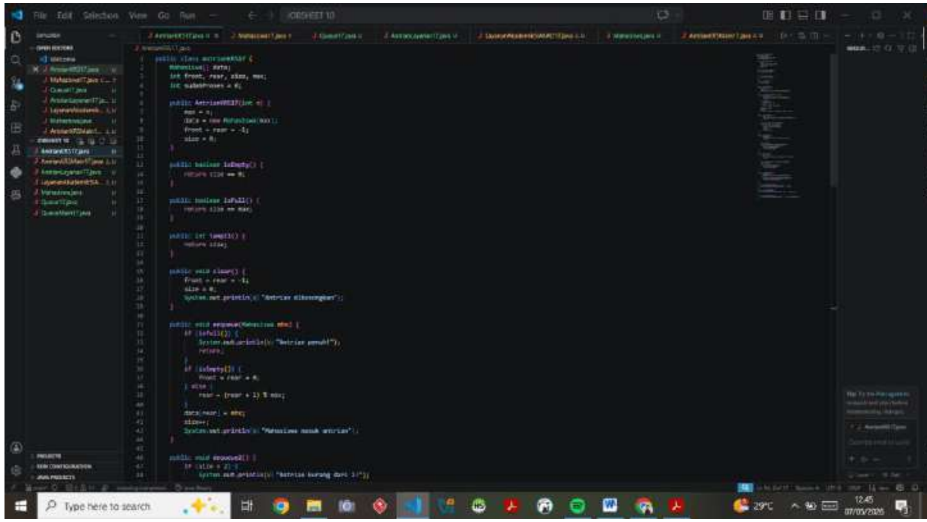
Switch:
case 6:
    antrian.lihatAkhir();
    break;

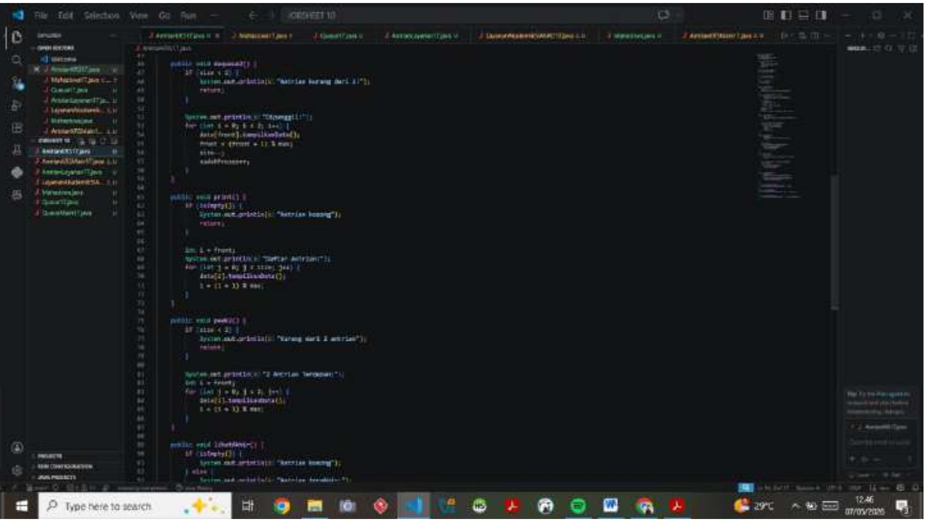
TUGAS – ANTRIAN KRS

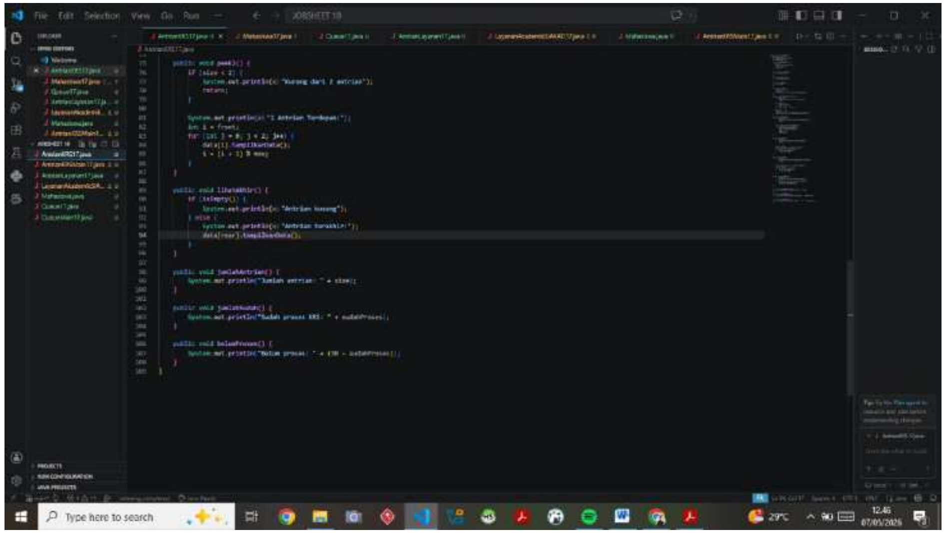
Class Mahasiswa



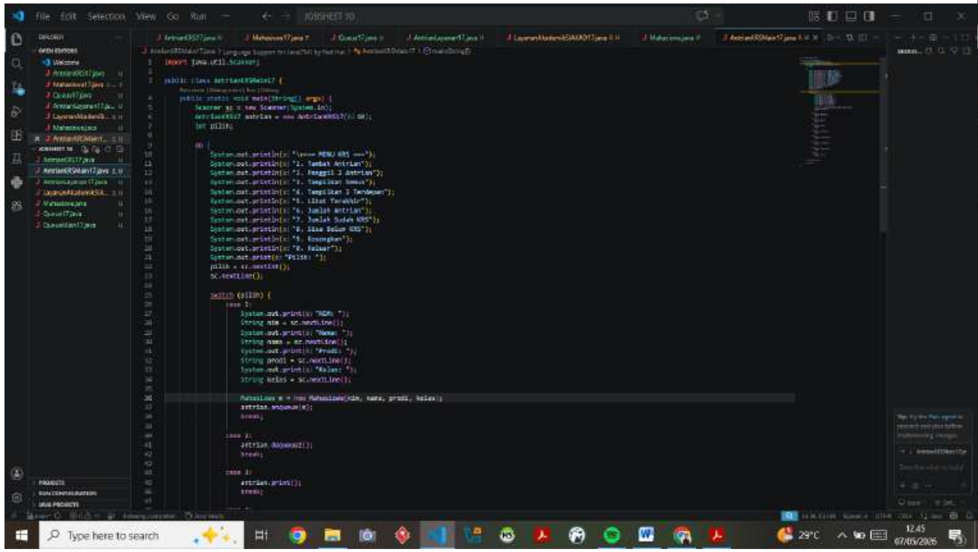
Class AntrianKRS

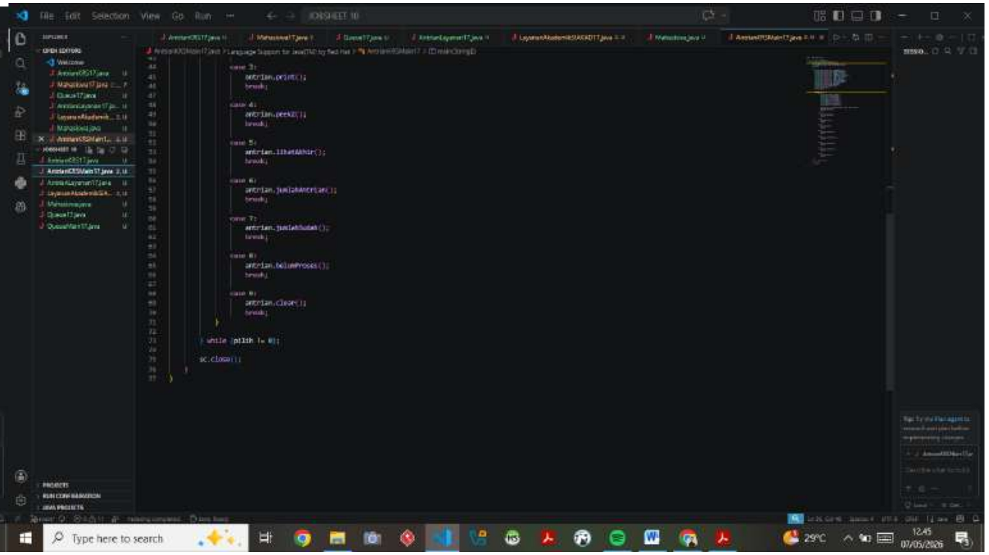






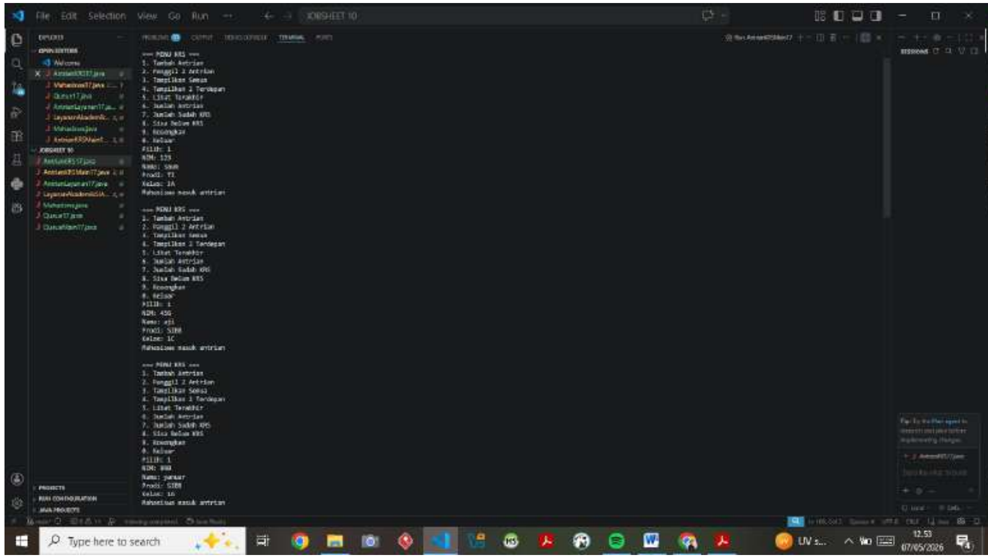
Class AntrianKRSMain

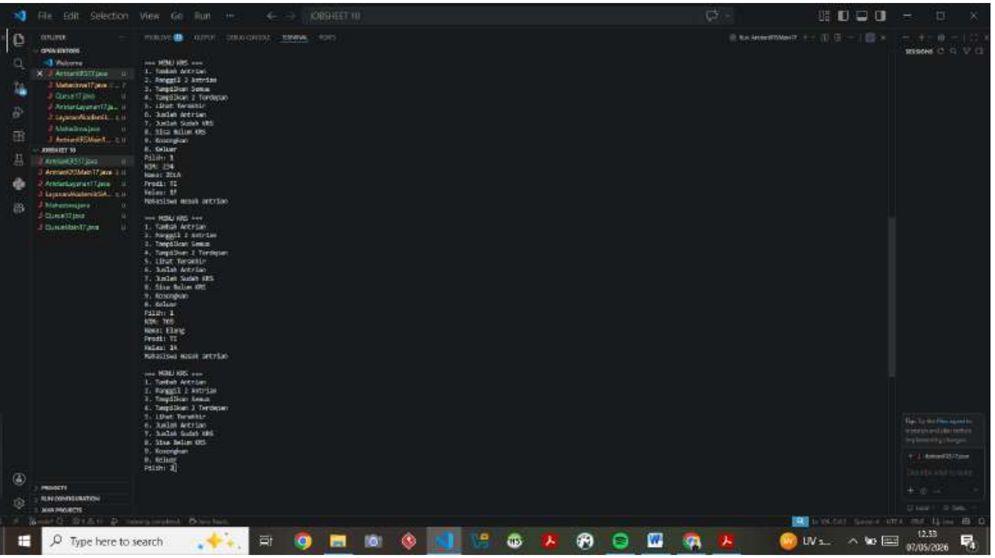




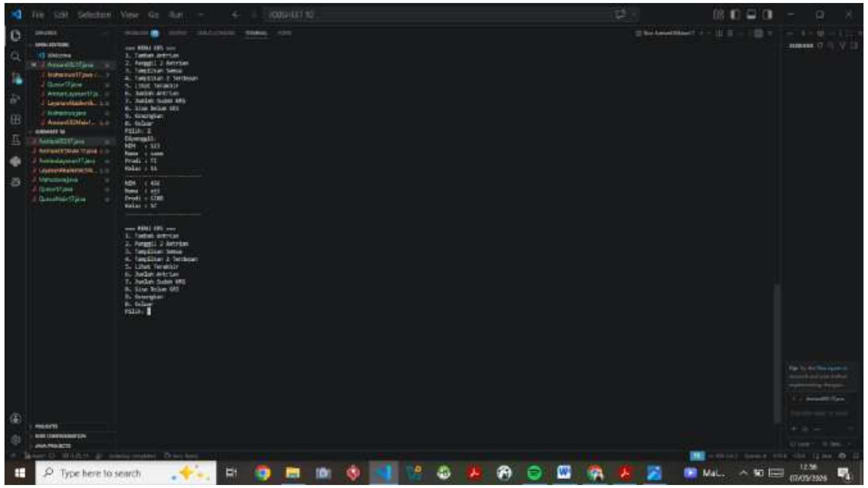
Hasil Run Program

Menu 1

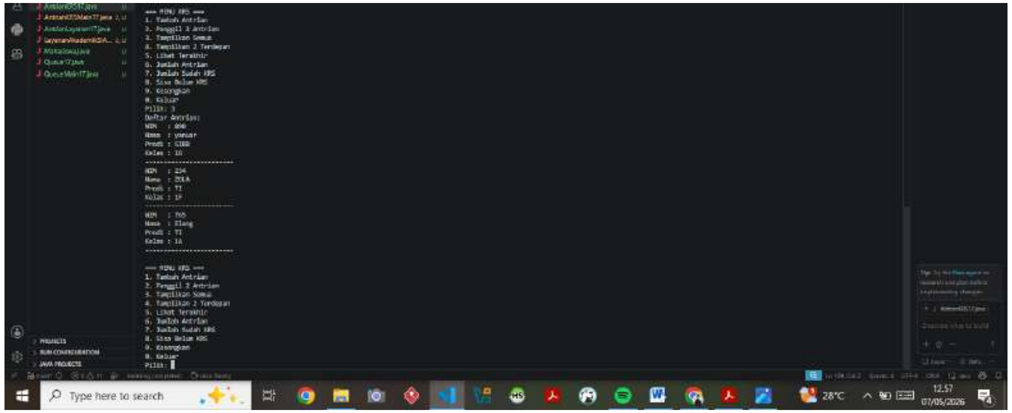




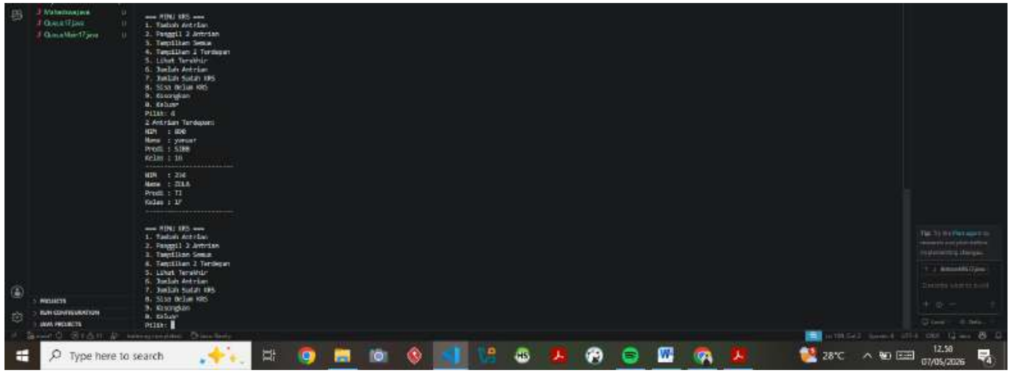
Menu 2



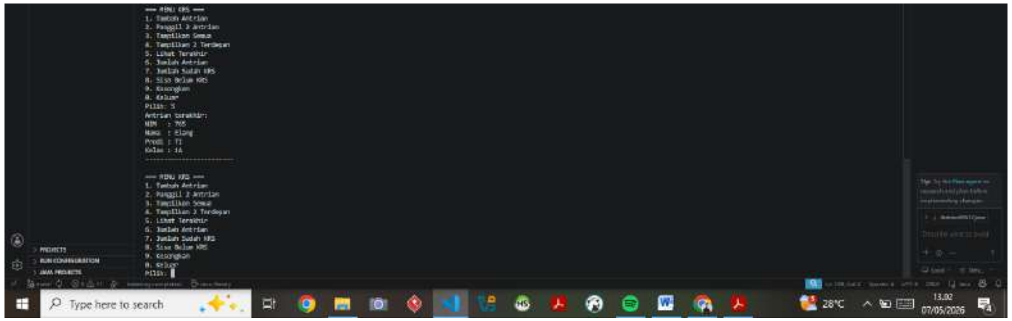
Menu 3



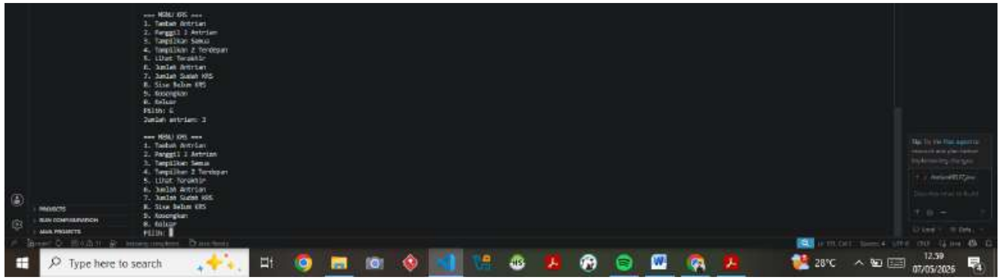
Menu 4



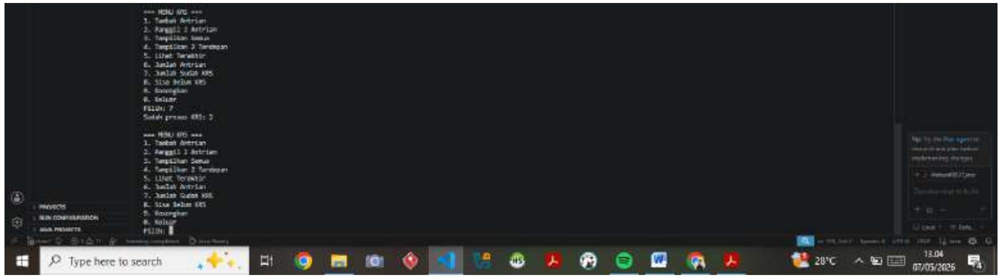
Menu 5



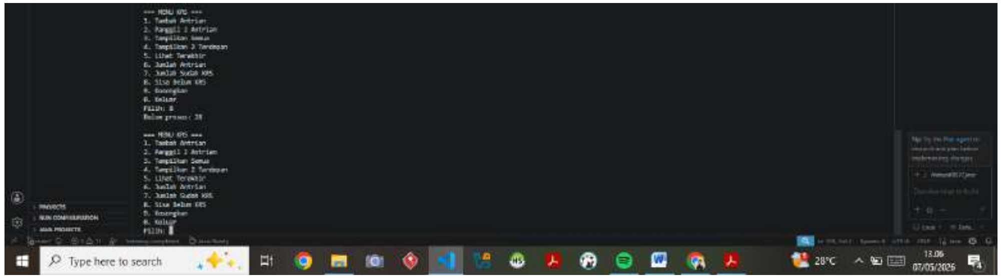
Menu 6



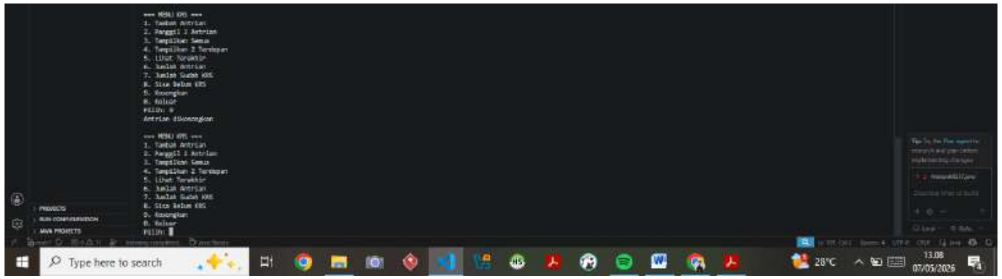
Menu 7



Menu 8



Menu 9



LinkGitHub: https://github.com/saumsaifulloh/AlgoritmaDanStrukturDataSem2/tree/main/ASD/PASD/JOBSHEET%2010